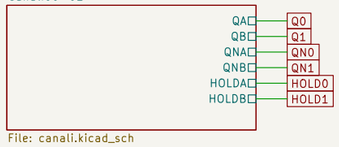
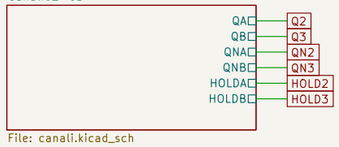
PianoMatrice — piano MuonScope 4x4 VBPW34S (passo 10 mm) sul bus MIKILAB REV5.1. Variabili: RF=${RF}, CF=${CF}, VBIAS=${VBIAS}. Vedi README.md.

Canali00–01



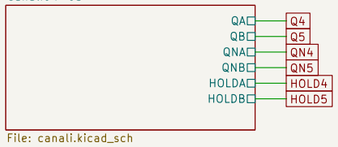
File: canali.kicad_sch

Canali02–03



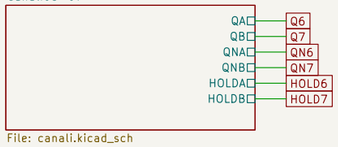
File: canali.kicad_sch

Canali04–05



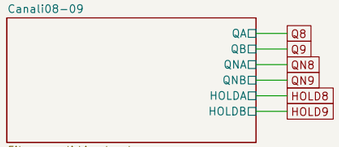
File: canali.kicad_sch

Canali06–07



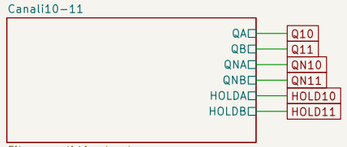
File: canali.kicad_sch

Canali08–09



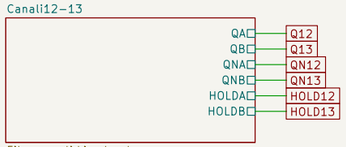
File: canali.kicad_sch

Canali10–11



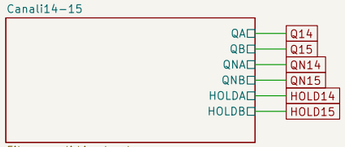
File: canali.kicad_sch

Canali12–13



File: canali.kicad_sch

Canali14–15



File: canali.kicad_sch

Acquisizione

File: acquisizione.kicad_sch

Bias

File: bias.kicad_sch

bus–pnp

File: bus–pnp.kicad_sch

Sheet: /
File: PianoMatrice.kicad_sch

Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 1/12

DUE CANALI (A, B) – foglio ripetuto 8 volte: canali 2k e 2k+1. Valori di retroazione da variabili di progetto: Rf = ${RF}, Cf = ${CF} (File > Impostazioni progetto > Variabili di testo).

Catena: VBPW34S –> CSA (OPA4140 A/B, +IN a 1V25_REF) –> shaper CR–RC invertente x50 tau 1 us (OPA4140 C/D) –> TLV3502 (soglia VTH) –> 74LVC74A (latch, CLR con CS_Y3) –> rilevatore di picco OPA2365 + BAS416 + 1 nF COG.

Iniezione di carica di test: 0.5 pF da TEST_INJ (gradino di -3.3 mV –> -1.6 fC). Rf2/Cf2 DNP in parallelo per prove. Trim di soglia per canale DNP (R tra VTH_x e GND).

U101
OPA4140AIPWR
1V25_REF
CSA_A
CSA_B
SUM_A
SUM_B
+5V
GND
+IN A
-IN A
+IN B
-IN B
+IN C
-IN C
+IN D
-IN D
V+
V–
OUTA
OUTB
OUTC
OUTD
CSO_A
CSO_B
SHO_A
SHO_B
R110
1k
DK_A
C114
1nF COG
SD_A
R116
1k
CP_A
R130
1k
DK_B
C134
1nF COG
SD_B
R136
1k
CP_B
C150
100nF
U102
TLV3502AIDGKR
CP_A
VTH_A
CP_B
VTH_B
+3V3
GND
CMP_A
CMP_B
C110
100nF/100V COG
R114
1k
VTH
R117
0R
C130
100nF/100V COG
R134
1k
R137
0R
C151
100nF
U103
SN74LVC74APWR
CMP_A
CMP_B
CS_Y3
1D
1CLK
1CLR
1PRE
2D
2CLK
2CLR
2PRE
VCC
1Q
1Q
2Q
2Q
QA
QNA
QB
QNB
R111
${RF}
R115
49.9k
R118
trim
R131
${RF}
R135
49.9k
R138
trim
C152
100nF
U104
OPA2365AIDR
PKI_A
HOLDA
PKI_B
HOLDB
PKO_A
PKO_B
D101
VBPW34S
D102
VBPW34S
C111
${CF} COG
C115
20pF COG
R119
1k
C131
${CF} COG
C135
20pF COG
R139
1k
C153
100nF
R112
Rf2
D11
BAS416
R132
Rf2
D13
BAS416
C154
100nF
C116
1nF COG
C112
Cf2
C132
Cf2
C136
1nF COG
Q110
BSS138
Q130
BSS138
PK_RST
C113
0.5pF COG
C133
0.5pF COG

Sheet: /Canali00–01/
File: canali.kicad_sch
Title:
Size: A2
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 2/12

DUE CANALI (A, B) — foglio ripetuto 8 volte: canali 2k e 2k+1. Valori di retroazione da variabili di progetto: Rf = ${RF}, Cf = ${CF} (File > Impostazioni progetto > Variabili di testo).

Catena: VBPW34S –> CSA (OPA4140 A/B, +IN a 1V25_REF) –> shaper CR–RC invertente x50 tau 1 us (OPA4140 C/D) –> TLV3502 (soglia VTH) –> 74LVC74A (latch, CLR con CS_Y3) –> rilevatore di picco OPA2365 + BAS416 + 1 nF COG.

Iniezione di carica di test: 0.5 pF da TEST_INJ (gradino di -3.3 mV –> -1.6 fC). Rf2/Cf2 DNP in parallelo per prove. Trim di soglia per canale DNP (R tra VTH_x e GND).

U201
OPA4140AIPWR
1V25_REF
CSA_A
CSA_B
SUM_A
SUM_B
+5V
GND
+IN A
-IN A
+IN B
-IN B
+IN C
-IN C
+IN D
-IN D
V+
V–
OUTA
OUTB
OUTC
OUTD
CSO_A
CSO_B
SHO_A
SHO_B
R210
1k
DK_A
C214
1nF COG
SO_A
R216
CP_A
R230
DK_B
C234
SO_B
R236
CP_B
C250
U202
TLV3502AIDGKR
CP_A
VTH_A
CP_B
VTH_B
+3V3
CMP_A
CMP_B
C210
100nF/100V COG
R214
SUM_A
R217
VTH_A
C230
R234
SUM_B
R237
VTH_B
C251
U203
SN74LVC74APWR
CMP_A
CMP_B
CS_Y3
1D
1CLK
1CLR
1PRE
2D
2CLK
2CLR
2PRE
VCC
1Q
1Q
2Q
2Q
QA
QNA
QB
QNB
R211
${RF}
CSO_A
R215
49.9k
SHO_A
R218
trim
C231
CSO_B
R235
SHO_B
R238
C252
U204
OPA2365AIDR
PKI_A
HOLDA
PKI_B
HOLDB
PKO_A
PKO_B
C211
${CF} COG
C215
20pF COG
R219
PKI_A
C231
${CF} COG
C235
20pF COG
R239
PKI_B
C253
D201
VBPW34S
D202
VBPW34S
R212
Rf2
Cf2
C212
D21
BAS416
D23
C254
100nF
C216
1nF COG
C236
Q210
BSS138
Q230
PK_RST
C213
0.5pF COG
C233
0.5pF COG

Sheet: /Canali02–03/
File: canali.kicad_sch
Title:
Size: A2
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 3/12

DUE CANALI (A, B) – foglio ripetuto 8 volte: canali 2k e 2k+1. Valori di retroazione da variabili di progetto: Rf = ${RF}, Cf = ${CF} (File > Impostazioni progetto > Variabili di testo).

Catena: VBPW34S –> CSA (OPA4140 A/B, +IN a 1V25_REF) –> shaper CR–RC invertente x50 tau 1 us (OPA4140 C/D) –> TLV3502 (soglia VTH) –> 74LVC74A (latch, CLR con CS_Y3) –> rilevatore di picco OPA2365 + BAS416 + 1 nF C0G.

Iniezione di carica di test: 0.5 pF da TEST_INJ (gradino di -3.3 mV –> -1.6 fC). Rf2/Cf2 DNP in parallelo per prove. Trim di soglia per canale DNP (R tra VTH_x e GND).

U301
OPA4140AIPWR
1V25_REF
CSA_A
1V25_REF
CSA_B
1V25_REF
SUM_A
1V25_REF
SUM_B
+5V
GND
+IN A
-IN A
+IN B
-IN B
+IN C
-IN C
+IN D
-IN D
V+
V–
OUTA
OUTB
OUTC
OUTD
CSO_A
CSO_B
SHO_A
SHO_B
U302
TLV3502AIDGKR
CP_A
VTH_A
CP_B
VTH_B
+3V3
GND
+IN A
-IN A
+IN B
-IN B
V+
V–
OUTA
OUTB
CMP_A
CMP_B
U303
SN74LVC74APWR
+3V3
CMP_A
CS_Y3
+3V3
+3V3
CMP_B
CS_Y3
+3V3
+3V3
GND
1D
1CLK
1CLR
1PRE
2D
2CLK
2CLR
2PRE
VCC
GND
1Q
1Q
2Q
2Q
QA
QNA
QB
QNB
U304
OPA2365AIDR
PKI_A
HOLDA
PKI_B
HOLDB
+3V3
GND
+IN A
-IN A
+IN B
-IN B
V+
V–
OUTA
OUTB
PKO_A
PKO_B
D301
VBPW34S
CSA_A
A
K
DK_A
D302
VBPW34S
CSA_B
A
K
DK_B
VBIAS_SH
R310
1k
DK_A
CSO_A
C314
1nF C0G
SO_A
SHO_A
R316
1k
CP_A
VBIAS_SH
R330
1k
DK_B
CSO_B
C334
1nF C0G
SO_B
SHO_B
R336
1k
CP_B
+5V
C350
100nF
GND
DK_A
C310
100nF/100V C0G
GND
SO_A
R314
1k
SUM_A
VTH
R317
0R
VTH_A
DK_B
C330
100nF/100V C0G
GND
SO_B
R334
1k
SUM_B
VTH
R337
0R
VTH_B
+5V
C351
100nF
GND
CSA_A
R311
${RF}
CSO_A
SUM_A
R315
49.9k
SHO_A
VTH_A
R318
trim
GND
CSA_B
R331
${RF}
CSO_B
SUM_B
R335
49.9k
SHO_B
VTH_B
R338
trim
GND
+3V3
C352
100nF
GND
CSA_A
C311
${CF} C0G
CSO_A
SUM_A
C315
20pF C0G
SHO_A
SHO_A
R319
1k
PKI_A
CSA_B
C331
${CF} C0G
CSO_B
SUM_B
C335
20pF C0G
SHO_B
SHO_B
R339
1k
PKI_B
+3V3
C353
100nF
GND
CSA_A
R312
Rf2
CSO_A
HOLDA
D31
PKO_A
BAS416
CSA_B
R332
Rf2
CSO_B
HOLDB
D33
PKO_B
BAS416
+3V3
C354
100nF
GND
CSA_A
C312
Cf2
CSO_A
HOLDA
C316
1nF C0G
GND
CSA_B
C332
Cf2
CSO_B
HOLDB
C336
1nF C0G
GND
PK_RST
Q310
BSS138
G
D
S
HOLDA
GND
PK_RST
Q330
BSS138
G
D
S
HOLDB
GND
TEST_INJ
C313
0.5pF C0G
CSA_A
TEST_INJ
C333
0.5pF C0G
CSA_B

Sheet: /Canali04–05/
File: canali.kicad_sch
Title:
Size: A2
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 4/12

DUE CANALI (A, B) – foglio ripetuto 8 volte: canali 2k e 2k+1. Valori di retroazione da variabili di progetto: Rf = ${RF}, Cf = ${CF} (File > Impostazioni progetto > Variabili di testo).

Catena: VBPW34S –> CSA (OPA4140 A/B, +IN a 1V25_REF) –> shaper CR–RC invertente x50 tau 1 us (OPA4140 C/D) –> TLV3502 (soglia VTH) –> 74LVC74A (latch, CLR con CS_Y3) –> rilevatore di picco OPA2365 + BAS416 + 1 nF C0G.

Iniezione di carica di test: 0.5 pF da TEST_INJ (gradino di -3.3 mV –> -1.6 fC). Rf2/Cf2 DNP in parallelo per prove. Trim di soglia per canale DNP (R tra VTH_x e GND).

U401
OPA4140AIPWR
1V25_REF
CSA_A
1V25_REF
CSA_B
1V25_REF
SUM_A
1V25_REF
SUM_B
+5V
GND
+IN A
-IN A
+IN B
-IN B
+IN C
-IN C
+IN D
-IN D
V+
V–
OUTA
OUTB
OUTC
OUTD
CSO_A
CSO_B
SHO_A
SHO_B
U402
TLV3502AIDGKR
CP_A
VTH_A
CP_B
VTH_B
+3V3
GND
+IN A
-IN A
+IN B
-IN B
V+
V–
OUTA
OUTB
CMP_A
CMP_B
U403
SN74LVC74APWR
+3V3
CMP_A
CS_Y3
+3V3
+3V3
CMP_B
CS_Y3
+3V3
+3V3
GND
1D
1CLK
1CLR
1PRE
2D
2CLK
2CLR
2PRE
VCC
GND
1Q
1Q
2Q
2Q
QA
QNA
QB
QNB
U404
OPA2365AIDR
PKI_A
HOLDA
PKI_B
HOLDB
+3V3
GND
+IN A
-IN A
+IN B
-IN B
V+
V–
OUTA
OUTB
PKO_A
PKO_B
D401
VBPW34S
CSA_A
A
K
DK_A
D402
VBPW34S
CSA_B
A
K
DK_B
VBIAS_SH
R410
1k
DK_A
CSO_A
C414
1nF C0G
SO_A
SHO_A
R416
1k
CP_A
VBIAS_SH
R430
1k
DK_B
CSO_B
C434
1nF C0G
SO_B
SHO_B
R436
1k
CP_B
+5V
C450
100nF
GND
DK_A
C410
100nF/100V C0G
GND
SO_A
R414
1k
SUM_A
VTH
R417
0R
VTH_A
DK_B
C430
100nF/100V C0G
GND
SO_B
R434
1k
SUM_B
VTH
R437
0R
VTH_B
+5V
C451
100nF
GND
CSA_A
R411
${RF}
CSO_A
SUM_A
R415
49.9k
SHO_A
VTH_A
R418
trim
GND
CSA_B
R431
${RF}
CSO_B
SUM_B
R435
49.9k
SHO_B
VTH_B
R438
trim
GND
+3V3
C452
100nF
GND
CSA_A
C411
${CF} C0G
CSO_A
SUM_A
C415
20pF C0G
SHO_A
SHO_A
R419
1k
PKI_A
CSA_B
C431
${CF} C0G
CSO_B
SUM_B
C435
20pF C0G
SHO_B
SHO_B
R439
1k
PKI_B
+3V3
C453
100nF
GND
CSA_A
R412
Rf2
CSO_A
HOLDA
D410
PKO_A
BAS416
CSA_B
R432
Rf2
CSO_B
HOLDB
D430
PKO_B
BAS416
+3V3
C454
100nF
GND
CSA_A
C412
Cf2
CSO_A
HOLDA
C416
1nF C0G
GND
CSA_B
C432
Cf2
CSO_B
HOLDB
C436
1nF C0G
GND
PK_RST
Q410
BSS138
G
D
S
HOLDA
GND
PK_RST
Q430
BSS138
G
D
S
HOLDB
GND
TEST_INJ
C413
0.5pF C0G
CSA_A
TEST_INJ
C433
0.5pF C0G
CSA_B

Sheet: /Canali06–07/
File: canali.kicad_sch
Title:
Size: A2
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 5/12

DUE CANALI (A, B) – foglio ripetuto 8 volte: canali 2k e 2k+1. Valori di retroazione da variabili di progetto: Rf = ${RF}, Cf = ${CF} (File > Impostazioni progetto > Variabili di testo).

Catena: VBPW34S –> CSA (OPA4140 A/B, +IN a 1V25_REF) –> shaper CR–RC invertente x50 tau 1 us (OPA4140 C/D) –> TLV3502 (soglia VTH) –> 74LVC74A (latch, CLR con CS_Y3) –> rilevatore di picco OPA2365 + BAS416 + 1 nF C0G.

Iniezione di carica di test: 0.5 pF da TEST_INJ (gradino di -3.3 mV –> -1.6 fC). Rf2/Cf2 DNP in parallelo per prove. Trim di soglia per canale DNP (R tra VTH_x e GND).

U501
OPA4140AIPWR
1V25_REF
CSA_A
CSA_B
SUM_A
SUM_B
+5V
GND
+IN A
-IN A
+IN B
-IN B
+IN C
-IN C
+IN D
-IN D
V+
V–
OUTA
OUTB
OUTC
OUTD
CSO_A
CSO_B
SHO_A
SHO_B
R510
1k
DK_A
C514
1nF C0G
SO_A
R516
1k
CP_A
R530
1k
DK_B
C534
1nF C0G
SO_B
R536
1k
CP_B
C550
100nF
U502
TLV3502AIDGKR
CP_A
VTH_A
CP_B
VTH_B
+3V3
CMP_A
CMP_B
C510
100nF/100V C0G
R514
1k
SUM_A
R517
0R
VTH_A
C530
100nF/100V C0G
R534
1k
SUM_B
R537
0R
VTH_B
C551
100nF
U503
SN74LVC74APWR
CMP_A
CMP_B
CS_Y3
1D
1CLK
1CLR
1PRE
1Q
1Q̄
2D
2CLK
2CLR
2PRE
VCC
GND
QA
QNA
QB
QNB
R511
${RF}
CSO_A
R515
49.9k
SHO_A
R518
trim
C531
${RF}
CSO_B
R535
49.9k
SHO_B
R538
trim
C552
100nF
U504
OPA2365AIDR
PKI_A
HOLDA
PKI_B
HOLDB
PKO_A
PKO_B
C511
${CF} C0G
C515
20pF C0G
R519
1k
PKI_A
C531
${CF} C0G
C535
20pF C0G
SHO_B
R539
1k
PKI_B
C553
100nF
D501
VBPW34S
CSA_A
DK_A
D502
VBPW34S
CSA_B
DK_B
R512
Rf2
CSO_A
D51
BAS416
HOLDA
PKO_A
C516
1nF C0G
C512
Cf2
Q510
BSS138
PK_RST
C513
0.5pF C0G
TEST_INJ
R532
Rf2
CSO_B
D53
BAS416
HOLDB
PKO_B
C554
100nF
C536
1nF C0G
C532
Cf2
Q530
BSS138
C533
0.5pF C0G

Sheet: /Canali08–09/
File: canali.kicad_sch
Title:
Size: A2
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 6/12

DUE CANALI (A, B) — foglio ripetuto 8 volte: canali 2k e 2k+1. Valori di retroazione da variabili di progetto: Rf = ${RF}, Cf = ${CF} (File > Impostazioni progetto > Variabili di testo).

Catena: VBPW34S –> CSA (OPA4140 A/B, +IN a 1V25_REF) –> shaper CR–RC invertente x50 tau 1 us (OPA4140 C/D) –> TLV3502 (soglia VTH) –> 74LVC74A (latch, CLR con CS_Y3) –> rilevatore di picco OPA2365 + BAS416 + 1 nF COG.

Iniezione di carica di test: 0.5 pF da TEST_INJ (gradino di -3.3 mV –> -1.6 fC). Rf2/Cf2 DNP in parallelo per prove. Trim di soglia per canale DNP (R tra VTH_x e GND).

U601
OPA4140AIPWR
1V25_REF
CSA_A
CSA_B
SUM_A
SUM_B
+5V
GND
+IN A
-IN A
+IN B
-IN B
+IN C
-IN C
+IN D
-IN D
V+
V–
OUTA
OUTB
OUTC
OUTD
CSO_A
CSO_B
SHO_A
SHO_B
R610
1k
DK_A
C614
1nF COG
SO_A
R616
1k
CP_A
R630
1k
DK_B
C634
1nF COG
SO_B
R636
1k
CP_B
C650
100nF
U602
TLV3502AIDGKR
CP_A
VTH_A
CP_B
VTH_B
+3V3
GND
+IN A
-IN A
+IN B
-IN B
V+
V–
OUTA
OUTB
CMP_A
CMP_B
C610
100nF/100V COG
R614
1k
VTH
R617
0R
C630
100nF/100V COG
R634
1k
R637
0R
C651
100nF
U603
SN74LVC74APWR
CMP_A
CS_Y3
+3V3
CMP_B
GND
1D
1CLK
1CLR
1PRE
2D
2CLK
2CLR
2PRE
VCC
GND
1Q
1Q
2Q
2Q
QA
QNA
QB
QNB
R611
${RF}
CSO_A
R615
49.9k
SHO_A
R618
trim
R631
${RF}
CSO_B
R635
49.9k
SHO_B
R638
trim
C652
100nF
U604
OPA2365AIDR
PKI_A
HOLDA
PKI_B
HOLDB
+3V3
GND
+IN A
-IN A
+IN B
-IN B
V+
V–
OUTA
OUTB
PKO_A
PKO_B
D601
VBPW34S
CSA_A
A
K
DK_A
C611
${CF} COG
C615
20pF COG
R619
1k
C631
${CF} COG
C635
20pF COG
R639
1k
C653
100nF
D602
VBPW34S
CSA_B
A
K
DK_B
R612
Rf2
D610
BAS416
HOLDA
PKO_A
R632
Rf2
D630
BAS416
HOLDB
PKO_B
C654
100nF
C616
1nF COG
C636
1nF COG
C612
Cf2
C632
Cf2
Q610
BSS138
PK_RST
G
D
S
HOLDA
GND
Q630
BSS138
PK_RST
G
D
S
HOLDB
GND
C613
0.5pF COG
TEST_INJ
C633
0.5pF COG
TEST_INJ

Sheet: /Canali10–11/
File: canali.kicad_sch
Title:
Size: A2
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 7/12

DUE CANALI (A, B) — foglio ripetuto 8 volte: canali 2k e 2k+1. Valori di retroazione da variabili di progetto: Rf = ${RF}, Cf = ${CF} (File > Impostazioni progetto > Variabili di testo).

Catena: VBPW34S –> CSA (OPA4140 A/B, +IN a 1V25_REF) –> shaper CR–RC invertente x50 tau 1 us (OPA4140 C/D) –> TLV3502 (soglia VTH) –> 74LVC74A (latch, CLR con CS_Y3) –> rilevatore di picco OPA2365 + BAS416 + 1 nF C0G.

Iniezione di carica di test: 0.5 pF da TEST_INJ (gradino di -3.3 mV –> -1.6 fC). Rf2/Cf2 DNP in parallelo per prove. Trim di soglia per canale DNP (R tra VTH_x e GND).

U701
OPA4140AIPWR
1V25_REF
CSA_A
1V25_REF
CSA_B
1V25_REF
SUM_A
1V25_REF
SUM_B
+5V
GND
+IN A
-IN A
+IN B
-IN B
+IN C
-IN C
+IN D
-IN D
V+
V–
OUTA
OUTB
OUTC
OUTD
CSO_A
CSO_B
SHO_A
SHO_B
U702
TLV3502AIDGKR
CP_A
VTH_A
CP_B
VTH_B
+3V3
GND
+IN A
-IN A
+IN B
-IN B
V+
V–
OUTA
OUTB
CMP_A
CMP_B
U703
SN74LVC74APWR
+3V3
CMP_A
CS_Y3
+3V3
+3V3
CMP_B
CS_Y3
+3V3
+3V3
GND
1D
1CLK
1CLR
1PRE
2D
2CLK
2CLR
2PRE
VCC
GND
1Q
1Q
2Q
2Q
QA
QNA
QB
QNB
U704
OPA2365AIDR
PKI_A
HOLDA
PKI_B
HOLDB
+3V3
GND
+IN A
-IN A
+IN B
-IN B
V+
V–
OUTA
OUTB
PKO_A
PKO_B
D701
VBPW34S
CSA_A
A
K
DK_A
D702
VBPW34S
CSA_B
A
K
DK_B
VBIAS_SH
R710
1k
DK_A
CSO_A
C714
1nF C0G
SO_A
SHO_A
R716
1k
CP_A
VBIAS_SH
R730
1k
DK_B
CSO_B
C734
1nF C0G
SO_B
SHO_B
R736
1k
CP_B
+5V
C750
100nF
GND
DK_A
C710
100nF/100V C0G
GND
SO_A
R714
1k
SUM_A
VTH
R717
0R
VTH_A
DK_B
C730
100nF/100V C0G
GND
SO_B
R734
1k
SUM_B
VTH
R737
0R
VTH_B
+5V
C751
100nF
GND
CSA_A
R711
${RF}
CSO_A
SUM_A
R715
49.9k
SHO_A
VTH_A
R718
trim
GND
CSA_B
R731
${RF}
CSO_B
SUM_B
R735
49.9k
SHO_B
VTH_B
R738
trim
GND
+3V3
C752
100nF
GND
CSA_A
C711
${CF} C0G
CSO_A
SUM_A
C715
20pF C0G
SHO_A
SHO_A
R719
1k
PKI_A
CSA_B
C731
${CF} C0G
CSO_B
SUM_B
C735
20pF C0G
SHO_B
SHO_B
R739
1k
PKI_B
+3V3
C753
100nF
GND
CSA_A
R712
Rf2
CSO_A
HOLDA
D710
PKO_A
BAS416
CSA_B
R732
Rf2
CSO_B
HOLDB
D730
PKO_B
BAS416
+3V3
C754
100nF
GND
CSA_A
C712
Cf2
CSO_A
HOLDA
C716
1nF C0G
GND
CSA_B
C732
Cf2
CSO_B
HOLDB
C736
1nF C0G
GND
PK_RST
Q710
BSS138
G
D
S
HOLDA
GND
PK_RST
Q730
BSS138
G
D
S
HOLDB
GND
TEST_INJ
C713
0.5pF C0G
CSA_A
TEST_INJ
C733
0.5pF C0G
CSA_B

Sheet: /Canali12–13/
File: canali.kicad_sch
Title:
Size: A2
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 8/12

DUE CANALI (A, B) — foglio ripetuto 8 volte: canali 2k e 2k+1. Valori di retroazione da variabili di progetto: Rf = ${RF}, Cf = ${CF} (File > Impostazioni progetto > Variabili di testo).

Catena: VBPW34S –> CSA (OPA4140 A/B, +IN a 1V25_REF) –> shaper CR–RC invertente x50 tau 1 us (OPA4140 C/D) –> TLV3502 (soglia VTH) –> 74LVC74A (latch, CLR con CS_Y3) –> rilevatore di picco OPA2365 + BAS416 + 1 nF C0G.

Iniezione di carica di test: 0.5 pF da TEST_INJ (gradino di -3.3 mV –> -1.6 fC). Rf2/Cf2 DNP in parallelo per prove. Trim di soglia per canale DNP (R tra VTH_x e GND).

U801
OPA4140AIPWR
1V25_REF
CSA_A
CSA_B
SUM_A
SUM_B
+5V
GND
+IN A
-IN A
+IN B
-IN B
+IN C
-IN C
+IN D
-IN D
V+
V–
OUTA
OUTB
OUTC
OUTD
CSO_A
CSO_B
SHO_A
SHO_B
R810
1k
DK_A
C814
1nF C0G
SD_A
R816
1k
CP_A
R830
1k
DK_B
C834
1nF C0G
SD_B
R836
1k
CP_B
C850
100nF
U802
TLV3502AIDGKR
CP_A
VTH_A
CP_B
VTH_B
+3V3
GND
CMP_A
CMP_B
C810
100nF/100V C0G
R814
1k
VTH
R817
0R
C830
100nF/100V C0G
R834
1k
R837
0R
C851
100nF
U803
SN74LVC74APWR
CMP_A
CMP_B
CS_Y3
1D
1CLK
1CLR
1PRE
1Q
1Q
2D
2CLK
2CLR
2PRE
VCC
2Q
2Q
QA
QNA
QB
QNB
R811
${RF}
R815
49.9k
R818
trim
R831
${RF}
R835
49.9k
R838
trim
C852
100nF
U804
OPA2365AIDR
PKI_A
HOLDA
PKI_B
HOLDB
PKO_A
PKO_B
C811
${CF} C0G
R819
1k
C831
${CF} C0G
C835
20pF C0G
R839
1k
C853
100nF
D801
VBPW34S
D802
VBPW34S
R812
Rf2
D810
BAS416
R832
Rf2
D830
BAS416
C854
100nF
C816
1nF C0G
C836
1nF C0G
C812
Cf2
C832
Cf2
Q810
BSS138
Q830
BSS138
PK_RST
C813
0.5pF C0G
C833
0.5pF C0G

Sheet: /Canali14–15/
File: canali.kicad_sch
Title:
Size: A2
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 9/12

MAPPA DEI COLPI: MCP23S17 U1001 (indirizzo Y5, CS_HITMAP) legge i 16 latch Q0–Q15 in una transazione. CONTROLLO: MCP23S17 U1002 (Y7, interrupt manager) — mux, test, FAULT, HIT_ANY.

AMPIEZZA: 2x TMUX1108 (8:1, 3 pA) sui 16 HOLD –> buffer OPA2365 –> 47R + 1 nF –> ADS8326 (Y0). Reset di latch e picchi: accesso all'indirizzo 3 (CS_Y3). Gradino di test: accesso all'indirizzo 4 con TEST_EN_N basso.

HIT_ANY = OR dei 16 latch (2x 74HC30 sui -Q + 74LVC1G32): un solo fronte per evento fino al reset.

U1001
MCP23S17-E/ML

U1002
MCP23S17-E/ML

U1003
TMUX1108PWR

U1004
TMUX1108PWR

U1005
OPA2365AIDR

U1006
ADS8326IDGKT

U1007
74HC30PW

U1008
74HC30PW

U1009
SN74LVC1G32DBVR

U1010
SN74LVC1G04DBVR

U1011
SN74LVC1G02DBVR

R1001
47R

R1002
100k

R1003
100R

R1004
10k

R1005
10k

R1006
100k

C1001
1nF C0G

C1002
100nF

C1003
100nF

C1004
100nF

C1005
100nF

C1006
100nF

C1007
100nF

C1008
100nF

C1009
100nF

C1010
100nF

C1011
100nF

TP1001
HIT_ANY

TP1002
MUX_D

TP1003
TEST_INJ

Sheet: /Acquisizione/
File: acquisizione.kicad_sch

Title:

Size: A2 | Date: | Rev:
KiCad E.D.A. 10.0.4 | Id: 10/12

BIAS: DAC61402 OUTA (V_BIAS nominale ${VBIAS}) -> shunt 10k (INA239, 125 pA/LSB) -> VBIAS_SH -> 16 filtri RC per diodo (1k + 100 nF C0G nei fogli dei canali).

SOGLIA: OUTB in range 0–5 V -> partitore 22k/10k -> VTH (0–1.56 V, ~0.4 mV/LSB), unica per il piano. TMP117 da piazzare al centro della matrice (lato sensori).

U1101
DAC61402RHBR

U1102
INA239AIDGSR

U1103
TMP117AIDRVR

R1101
100R

C1106
100nF/100V C0G

R1103
22k

C1108
100nF

C1101
1uF/100V

R1104
10k

C1109
100nF

C1102
1uF/100V

C1107
100nF

C1110
100nF

D1101
BAT54S

C1103
100nF/100V

R1105
10k

C1111
100nF

C1104
470pF/100V

TP1101
VBIAS_SH

C1112
100nF

C1105
470pF/100V

TP1102
VTH

R1102
10k 0.1%

Sheet: /Bias/	
File: bias.kicad_sch	
Title:	
Size: A2	Date:
KiCad E.D.A. 10.0.4	Rev: Id: 11/12

BUS MIKILAB REV5.1 – MAIN 28 pin, come il PianoTest, Y0 ADC, Y1 DAC, Y2 INA, Y3 reset latch e picchi, Y4 gradino di test, Y5 mappa dei colpi (MCP), Y6 EEPROM ID, Y7 MCP di controllo.

HIT# (pin 28): HIT_ANY –> monostabile 74LVC1G123 (2k + 27 pF, ~170–200 ns) –> 74LVC1G06 open-drain. ALERT#: INTA dell'MCP di controllo via 74LVC1G07.

J1201
J_IN MAIN
GND 1
+5V 2
+3V3 3
GND 4
2V5_REF 5
1V25_REF 6
GND_A 7
VHV_36V 8
VHV_36V_2 9
GND 10
BUS_SCLK 11
BUS_MOSI 12
BUS_MISO 13
I2C_SDA 14
I2C_SCL 15
GND 16
BUS_CS_MASTER 17
BUS_ADDR0 18
BUS_ADDR1 19
BUS_ADDR2 20
BUS_PREFIX0 21
BUS_PREFIX1 22
BUS_PREFIX2 23
BUS_ALERT_N 24
ID_IN0 25
ID_IN1 26
ID_IN2 27
BUS_HIT_N 28
MH
GND

J1202
J_OUT MAIN
GND 1
+5V 2
+3V3 3
GND 4
2V5_REF 5
1V25_REF 6
GND_A 7
VHV_36V 8
VHV_36V_2 9
GND 10
BUS_SCLK 11
BUS_MOSI 12
BUS_MISO 13
I2C_SDA 14
I2C_SCL 15
GND 16
BUS_CS_MASTER 17
BUS_ADDR0 18
BUS_ADDR1 19
BUS_ADDR2 20
BUS_PREFIX0 21
BUS_PREFIX1 22
BUS_PREFIX2 23
BUS_ALERT_N 24
ID_OUT0 25
ID_OUT1 26
ID_OUT2 27
BUS_HIT_N 28
MH
GND

U1201
74HC283
ID_IN0 5
ID_IN1 3
ID_IN2 14
GND 12
GND 6
GND 2
GND 15
GND 11
GND 7
+3V3 16
+3V3 8
GND
A1 A2 A3 A4 B1 B2 B3 B4 CIN VCC GND
S1 4 ID_OUT0
S2 1 ID_OUT1
S3 13 ID_OUT2
S4 10
COUT 9

U1202
74HC85
BUS_PREFIX0 10
BUS_PREFIX1 12
BUS_PREFIX2 13
GND 15
ID_IN0 9
ID_IN1 11
ID_IN2 14
GND 1
GND 2
+3V3 3
GND 4
+3V3 16
GND 8
A0 A1 A2 A3 B0 B1 B2 B3 IA<B IA=B IA>B VCC GND
QA>B 5 AGTB
QA=B 6 PL_SEL
QA<B 7 ALTB

U1203
74HC138
BUS_ADDR0 1
BUS_ADDR1 2
BUS_ADDR2 3
BUS_CS_MASTER 4
GND 5
PL_SEL 6
+3V3 16
GND 8
A0 A1 A2 E1 E2 E3 VCC GND
Y0 15 CS_ADC
Y1 14 CS_DAC
Y2 13 CS_INA
Y3 12 CS_Y3
Y4 11 CS_Y4
Y5 10 CS_HITMAP
Y6 9 CS_EEP
Y7 7 CS_MCP

U1204
SN74LVC2G17DBVR
BUS_SCLK 1
BUS_MOSI 3
+3V3 5
GND 2
1A 2A VCC GND
1Y 6 SCLK_L
2Y 4 MOSI_L

U1205
SN74LVC1G125DBVR
MISO_L 2
MISO_OE_N 1
+3V3 5
GND 3
A OE VCC GND
Y 4 BUS_MISO

U1206
SN74LVC1G332DBVR
BUS_CS_MASTER 1
AGTB 3
ALTB 6
+3V3 5
GND 2
A B C VCC GND
Y 4 MISO_OE_N

U1207
SN74LVC1G07DBVR
MCP_INTA 2
+3V3 5
GND 3
A NC VCC GND
Y 4 BUS_ALERT_N

U1209
SN74LVC1G123DCUR
GND 1
HIT_ANY 2
+3V3 3
+3V3 8
GND 4
A B CLR VCC GND
Q 5 HIT_PULSE
Rext/Cext 7 MONO_RC
Cext 6 MONO_C

U1210
SN74LVC1G06DBVR
HIT_PULSE 2
+3V3 5
GND 3
A NC VCC GND
Y 4 BUS_HIT_N

U1208
25AA02E48T-I/SN
CS_EEP 1
SCLK_L 6
MOSI_L 5
+3V3 3
+3V3 7
+3V3 8
GND 4
CS SCK SI WP HOLD VCC VSS
SO 2 MISO_L

GND_A
R1201
0R
GND

MONO_RC
C1215
27pF C0G
MONO_C

+3V3
C1201
100nF
GND

+3V3
C1208
100nF
GND

2V5_REF
C1216
1uF
GND

BUS_SCLK

+3V3
R1202
100k
BUS_CS_MASTER

+3V3
C1202
100nF
GND

+3V3
C1209
100nF
GND

1V25_REF
C1217
1uF
GND

BUS_MOSI

MISO_L
R1203
100k
GND

+3V3
C1203
100nF
GND

+3V3
C1210
100nF
GND

BUS_MISO

ID_IN0
R1204
100k
GND

+3V3
C1204
100nF
GND

+3V3
C1211
10uF
GND

I2C_SDA

ID_IN1
R1205
100k
GND

+3V3
C1205
100nF
GND

+5V
C1212
10uF
GND

I2C_SCL

BUS_CS_MASTER

ID_IN2
R1206
100k
GND

+3V3
C1206
100nF
GND

VHV_36V
C1213
100nF/100V
GND

BUS_ADDR0

+3V3
R1207
2k
MONO_RC

+3V3
C1207
100nF
GND

VHV_36V_2
C1214
100nF/100V
GND

BUS_ADDR1

BUS_ADDR2

Sheet: /bus-pnp/
File: bus-pnp.kicad_sch
Title:
Size: A2 | Date: | Rev:
KiCad E.D.A. 10.0.4 | Id: 12/12